



WHITE PAPER

Launch Your DataOps Journey with the DataOps Maturity Model

MOST ENTERPRISE COMPANIES recognize the need to be data-driven, yet 60% of data projects fail to move past preliminary stages, and 87% of data science projects never make it to production. More surprisingly the number of data-driven companies has actually fallen from 37% to 31% since 2017, despite increased investment, according to Gartner.

WHY? Becoming data-driven is hard. Data teams are caught between the competing demands of data consumers, data providers, and supporting teams. Typically, data consumers live in an “Amazon world” and expect trusted, original insight on-demand. Yet data providers often send inaccurate, late, or error-prone data sets. The flawless collaboration demanded of stakeholders often just isn’t there.

Taken together, the need to manage complex toolchains and data, as well as collaborate with other organizations, roles, locations, and data centers, saps the data team’s time. In fact, most data teams spend more time fixing errors and addressing operational issues than innovating and providing business value. According to Gartner, only 22% of a data team’s time is spent on new initiatives and innovation (Figure 1). As a result, many data teams are not meeting expectations, or worse, are beaten down and disempowered.

March 2020 Gartner Survey:
“Data Management Struggles to Balance Innovation and Control”

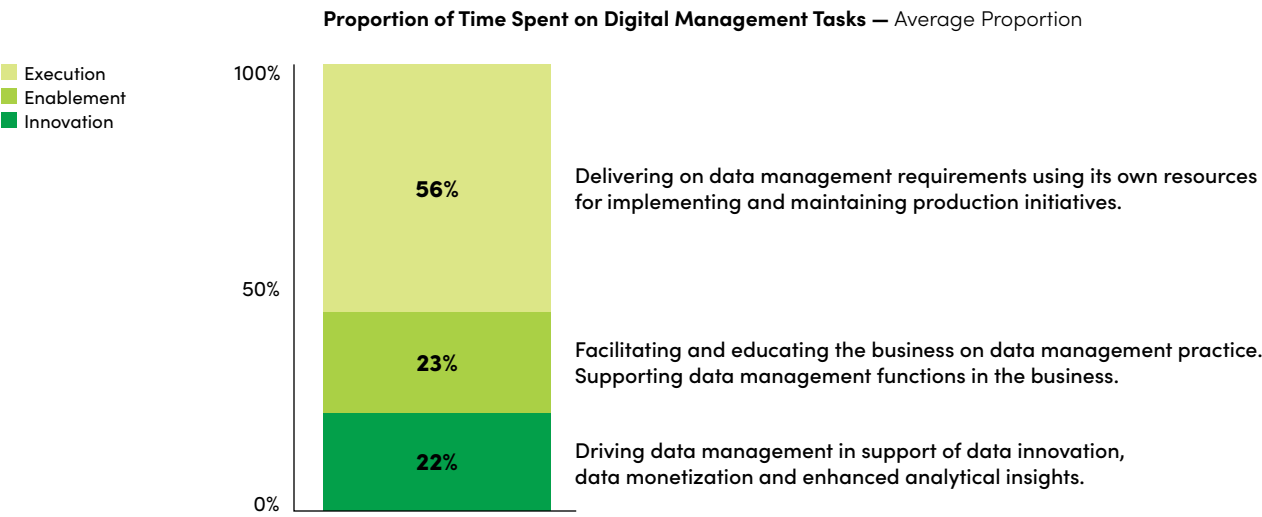


FIGURE 1: Only 22% of a data team’s time is spent on new initiatives and innovation.

In data analytics, DataOps provides the path forward. Research shows that “organizations that adopt a DevOps- and DataOps-based approach are more successful in implementing end-to-end, reliable, robust, scalable and repeatable solutions,” says Gartner’s Sumit Pal. (Gartner, November 2018)

WHAT IS DATAOPS?

DataOps is a set of technical practices, cultural norms, and architectures that enables:

- Rapid experimentation and innovation for the fastest delivery of new insights to customers
- Low error rates
- Collaboration across complex sets of people, technology, and environments
- Clear measurement and monitoring of results

DataOps draws on the principles of Agile, DevOps, and Lean manufacturing to transform data processes. Supported by automation, it puts the focus on the underlying systems and managing the ‘data factory.’ Companies that implement DataOps realize multiple, simultaneous benefits. They:

- Experience orders of magnitude improvements in cycle time. They are able to deploy new analytics quickly and confidently, often delivering in hours/days instead of weeks/months.
- Lower or even eliminate costly and embarrassing errors, enabling organizations to build a strong culture of trust with their data customers.
- Dramatically increase productivity. Better intra- and inter-team collaboration means less time spent on meetings and bureaucracy and more on innovation.

A MATURITY MODEL CAN HELP YOU GET STARTED

Because DataOps impacts your end-to-end analytic lifecycle, implementing DataOps can feel overwhelming. Even though a majority of respondents in a 2020 Seagate/ IDC survey said that DataOps was “very” or “extremely” important, only 10% have implemented DataOps fully across the enterprise. Success requires a mindset shift and most companies struggle with where to begin and how to even make modest progress towards their goals. A DataOps Maturity Model can be an incredibly useful tool to help organizations understand where they are today and how to get where they need to go.

WHAT IS IMPORTANT IN DATAOPS?

To begin a DataOps initiative, it is first important to understand what is important (and what isn't) for DataOps success. DataOps requires a focus on the state of your data operations and processes, not the next new feature or tool. Typically, data teams can spend far too much time worrying about data types (e.g., batch, streaming, big, small, structured, unstructured), database types (e.g., hadoop, Spark, graph, NoSQL, object stores), data tools (e.g., ETL, BI, data science, data prep, catalog, etc), or specific design paradigms (e.g., lakes, warehouses, ML models, etc).

DataOps employs automated orchestration to simplify complex toolchains, environments, and team collaboration, so that the data team can quickly and continuously deliver high-quality, error-free insight. To implement DataOps, organizations need to prioritize improvements in the six following areas (Figure 2).

- *Error Rates*
- *Cycle Time*
- *Collaboration*
- *Measurement*
- *Team Culture*
- *Customer Happiness*

Each of these areas can be improved by implementing core DataOps capabilities such as automated testing and monitoring, toolchain orchestration, version control, sandbox creation and management, and continuous deployment. Figure 3 highlights areas where DataOps capabilities can help you address key business constraints.

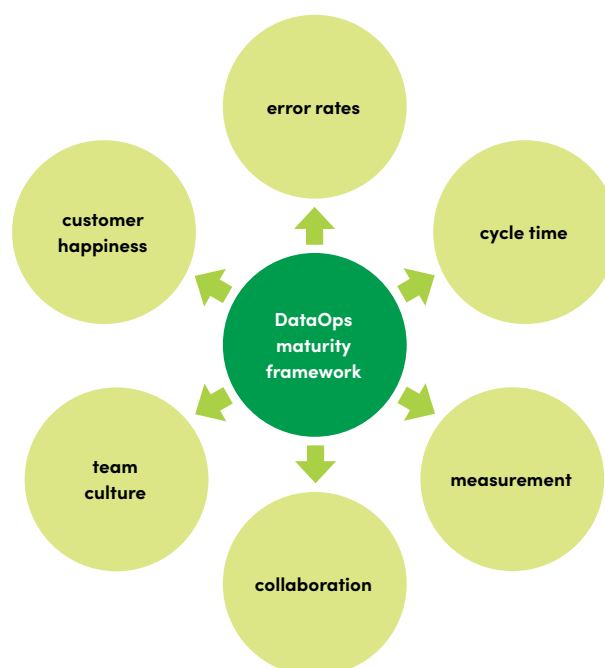
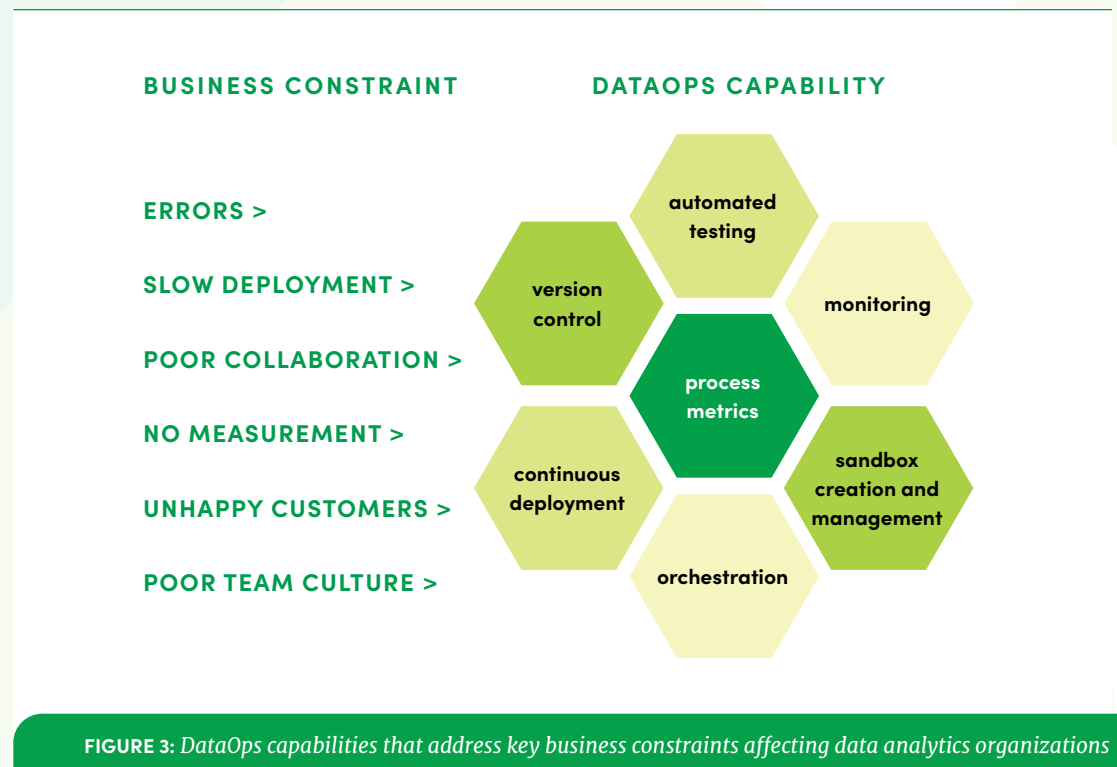


FIGURE 2: The 6 dimensions of DataOps Maturity



A DATAOPS MATURITY MODEL

Maturity models are commonly used to measure an organization's ability to continuously improve in a particular discipline. This document outlines a maturity model measurement approach for building, monitoring, and deploying data and analytics according to DataOps principles. With this model, teams can understand where they are today, and what needs to be done to move up the curve.

The model provides a structure for reviewing your organization's capabilities across the six different DataOps dimensions. Results will enable you to customize strategies to get started or improve. A robust DataOps program will be optimized across all the dimensions.

THE SIX PRIMARY DATAOPS DIMENSIONS

A DataOps Maturity Assessment asks questions across the six categories that form the dimensions of the DataOps Maturity Model. Along each dimension, progress toward maturity can be categorized as:

Level 5 Optimized: Focus on continuous improvement and change.

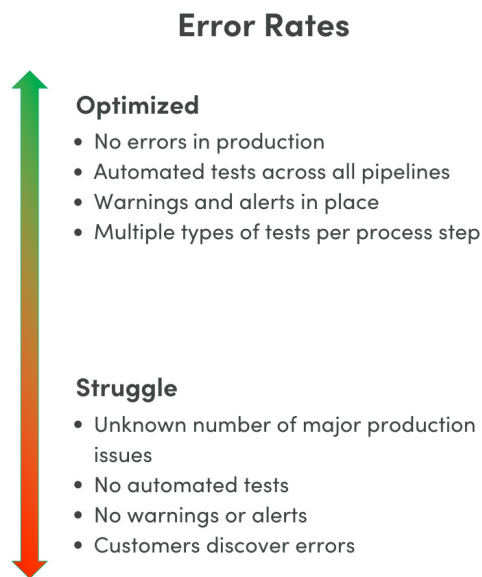
Level 4 Quantitative: Processes are measured and controlled.

Level 3 Consistent: Automated processes are being applied across the entire data analytic development lifecycle.

Level 2 Basic: Processes are documented and partly automated.

Level 1 Struggle: Processes are unrepeatable, poorly controlled, manual, and reactive.

FIGURE 4: Progress toward DataOps maturity for Error Rates.



Cycle Time

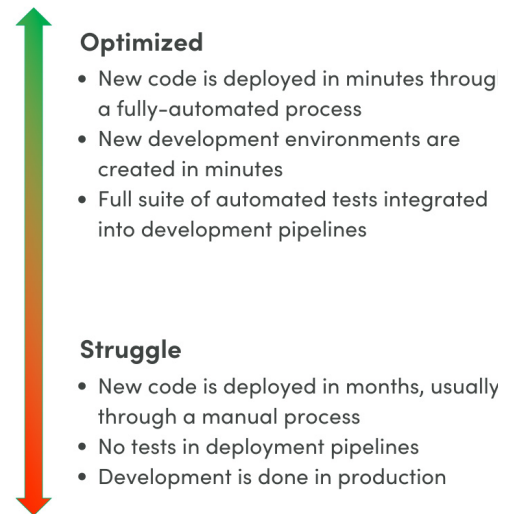


FIGURE 5: Progress toward DataOps maturity for Cycle Time.

Every company is on a journey toward achieving excellence and will have strengths and weaknesses. Just because an organization is large, does not mean that it is excellent. In fact, the flaws in a process or methodology become particularly noticeable when a team grows. Low ratings on any dimension should not be viewed as a negative, but instead as an opportunity for improvement.

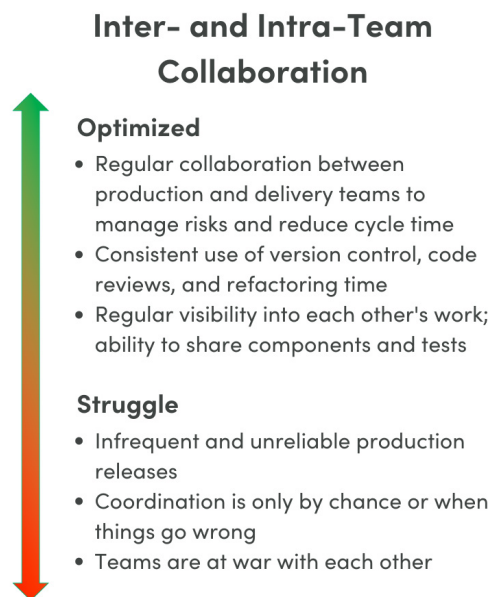
PRODUCTION ERROR RATES

Organizations that follow DataOps principles typically have less than one error per year. That is orders of magnitude better than the industry norm. In a recent DataOps survey, only 3% of the companies surveyed approached that level of quality. Eighty percent of companies surveyed reported three or more errors per month. Thirty percent of respondents reported more than 11 errors per month.

To reduce the level of errors, robust DataOps programs use [automated testing, monitoring, and orchestration in their production pipelines](#). Inspired by statistical process control, they will have tests running in production across all pipelines, sources, and tools, multiple types of tests per process step, and error alerts in place.

In contrast, teams that struggle will have no automated tests in production. This results in costly and embarrassing errors, often discovered by customers. (Figure 4)

FIGURE 6: Progress toward DataOps maturity for Collaboration.



Measurement



FIGURE 7: Progress toward DataOps maturity for Measurement.

DEPLOYMENT CYCLE TIME

Many organizations experience lengthy cycle times for creating analytic environments or deploying new [analytics that run weeks and months](#). This is often due to manual processes with little to no automation or automated testing in place. In the worst case scenario, development work is done in the production technical environment that also hosts live data operations.

Deployment cycle time can be shortened through a strong program of testing, deployment automation, and environment management. Optimized DataOps programs can deploy new analytics and create new development environments in hours, or even minutes. In DataOps, [error-free automated deployment](#) is realized through a full suite of tests. (Figure 5)

WELL-COORDINATED INTER- AND INTRA-TEAM COLLABORATION

Organizations with an optimized DataOps program have high levels of inter- and intra-team collaboration. In these organizations, production and development teams regularly collaborate to reduce risk, speed cycle time, and achieve overall greater productivity. They are able to share assets between teams and have visibility into each other's work. These teams also leverage environments, version control, and code review processes for successful collaboration.

On the contrary, organizations that struggle tend to have unreliable processes, chance meetings (and often yelling when things go wrong). Analytic and line-of-business teams are often at war with each other. (Figure 6)

FIGURE 8: Progress toward DataOps maturity for Team Culture..



Happy Customers



FIGURE 9: Progress toward DataOps maturity for Happy Customers.

SUCCESS AND FAILURE MEASURED

You can't improve what you don't measure, yet it is surprising how many data analytics teams don't measure their own processes. Optimized DataOps teams continuously measure success and failure through detailed process analytics on errors, deployment speed, and team productivity. Metrics are regularly shared and reviewed with the team and internal customers, with a focus on improvement. Conversely, teams that struggle don't track metrics or create reports, or worse, don't collect any data at all. (Figure 7)

TEAM CULTURE

DataOps draws upon the principles of Agile and Lean manufacturing to transform processes that manage data on its journey toward value creation. Successful DataOps teams follow Agile principles which are a strong part of the overall company culture. These organizations are focused on continuous learning and optimization and errors are viewed as an opportunity for improvement.

On the opposite end of the spectrum are companies that follow waterfall principles. Errors go undiscovered or are hidden and blame is passed around when things go wrong. (Figure 8)

HAPPY CUSTOMERS

Customers won't adopt analytics they don't trust. At the end of the day, delivering trusted, timely insight is a critical measure of success. Best-in-class data teams will respond to customer requests within hours and always provide timely, useful insight. Teams that struggle are often 'too busy' to respond. Customers begin to look elsewhere for insight – the death knell for a data analytics team. (Figure 9)

WHAT DOES GOOD LOOK LIKE?

No company will initially excel across all six dimensions. Initially, typical results will look like Figure 10. However, the results of your maturity assessment will help your organization plan a roadmap for success. The goal of a DataOps maturity assessment is to provoke a discussion across your organization, to identify areas for improvement, and to guide investment in the processes and tools that can help.

The good news is that success in one particular dimension does not have to be traded-off against others. In organizations that don't practice DataOps, this is a common practice. They often trade speed for quality (or vice versa). For example, in order to reduce fear and uncertainty over errors, a team may establish practices, like documentation, checks and balances, and lots of meetings, that lengthen their cycle time and reduce productivity. With DataOps practices in place, you can excel in both speed and quality. Best-in-class data organizations do well across the board, leading to overall greater productivity and lower costs.

DataOps Maturity Model Score

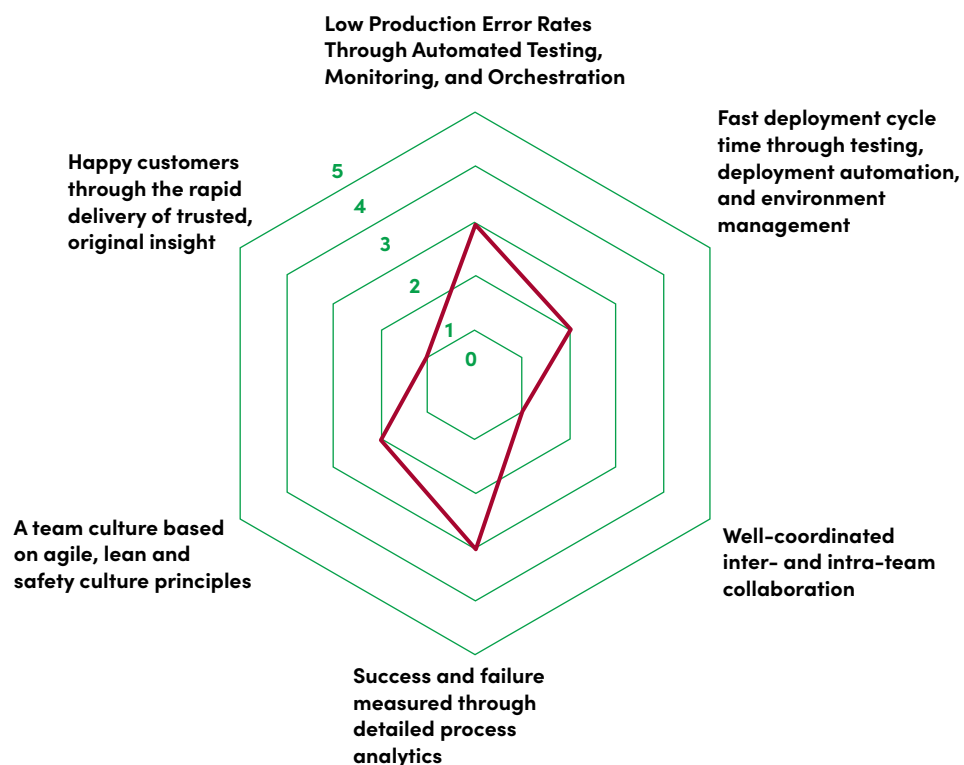


FIGURE 10: Typical DataOps Maturity Model results

By focusing on the right areas, a data team can start to look more like Bristol Myers Squibb (formerly Celgene), a company that is now several years into their DataOps journey (Figure 11). This team initially overcame obstacles that prevented analytics responsiveness and quality. Data was organized in silos – using a variety of technologies and isolated platforms. Without the right processes and tools in place, the data engineering and analytics teams spent a majority of their time on data engineering and pipeline maintenance. This distracted them from their main mission – producing analytic insights that help the business attain its objectives.

After implementing DataOps, they now achieve excellence across all critical dimensions:

- Very, very few errors or missed SLAs
- Weekly cycle time of new changes/features/data
- Detailed process metrics
- Agile culture
- High inter- and intra-team coordination
- High customer satisfaction

BMS DataOps Maturity Model Score

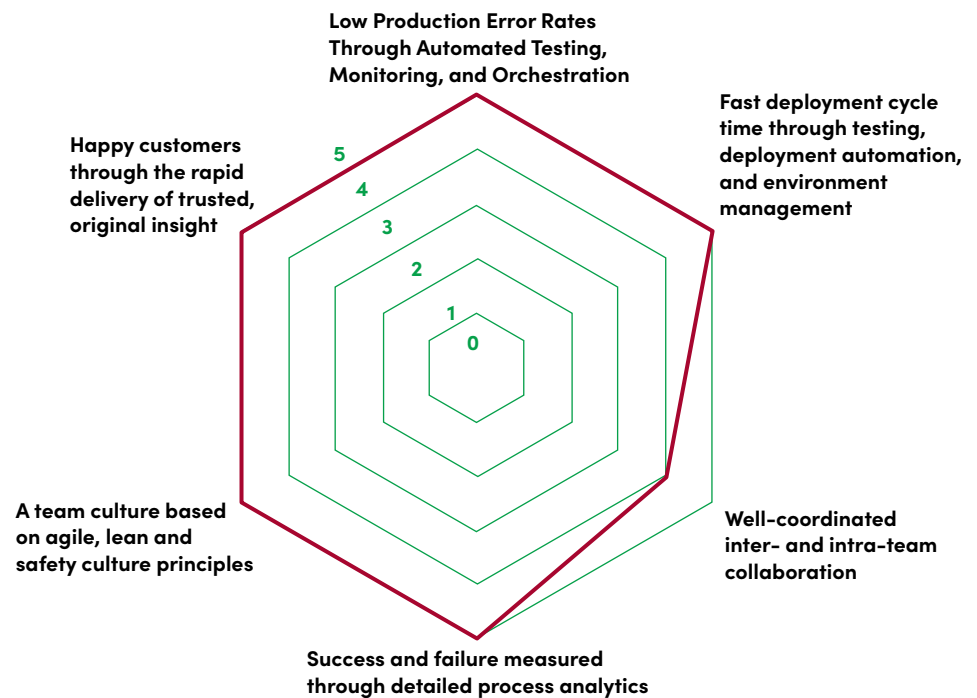


FIGURE 11: BMS is optimized across all dimensions of the DataOps maturity model

THE JOURNEY TO EXCELLENCE

As most organizations come to recognize the benefits of a DataOps program, adoption is often a no-brainer. DataOps provides the foundation for analytic excellence. It streamlines the development of new analytics, shortens cycle time, and automates the data analytic pipeline, freeing the team to focus on value-add activities. It also controls the quality of the data flowing through the pipeline so users can trust their data. With DataOps in place, the team is productive, responsive, and efficient.

Because implementation of DataOps requires a mindset shift, one of the biggest challenges becomes where and how to start. The DataOps Maturity Model provides a quick objective way for organizations to assess the maturity of their DataOps initiative and breaks down the critical elements of a DataOps program into concrete, actionable areas for improvement.

Ready to get started? Please reach out to us at info@datakitchen.io to take the assessment.